

Perilesional MRE forward-and-inverse problem

3D scalar Helmholtz and vector Navier on the Sobh–Ehman phantoms

A step-by-step specification for the computational team

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Purpose.

This note fully specifies the 3D time-harmonic MRE forward simulation used to generate the shear-wave displacement field $u(\mathbf{x})$ inside a water-balloon phantom, together with the two inversion pipelines applied to u . Everything a code reviewer or a re-implementer needs to reproduce the runs is here: geometry, boundary conditions, material model, driver, discretisation, inversion recipes, and the numerical scales that make the run well-posed.

1 Physical picture

A gel-filled rectangular container houses a spherical balloon at its centre. The balloon has been inflated with water to 250 mL (radius $a = 3.908$ cm) from a stress-free casting state at 50 mL (radius $A = 2.285$ cm). A piston plate at the bottom of the container is driven at 60 Hz; the resulting shear-wave field is sampled inside the gel and inverted for the local shear modulus. This mimics the MRE experiment reported by Yin et al. (2026) as reproduced quantitatively by *paper_lesion_expansion.pdf*.

2 Governing equation

We solve the scalar Helmholtz equation for the complex tangential displacement $u(\mathbf{x}) \in \mathbb{C}$ at fixed angular frequency $\omega = 2\pi f$ with $f = 60$ Hz:

$$\rho \omega^2 u(\mathbf{x}) = \nabla \cdot (G^*(\mathbf{x}) \nabla u(\mathbf{x})) \quad \mathbf{x} \in \Omega, \quad (1)$$

where the complex-valued shear modulus incorporates hysteretic damping:

$$G^*(\mathbf{x}) = G(\mathbf{x}) (1 + i \xi), \quad \xi = 0.05 \quad (\Rightarrow Q \approx 10). \quad (2)$$

Physical constants:

Symbol	Value	Meaning
ρ	1000 kg m^{-3}	gel density
ω	$2\pi \cdot 60 \text{ rad s}^{-1}$	angular frequency
ξ	0.05	hysteretic damping ($Q \approx 10$)

Rationale for the scalar reduction: the acoustoelastic tangential modulus $\mu_{\text{app},\theta\theta}$ appears in the scalar equation governing the ring-tangential polarisation, which is what TSM MIP isolates. Full vector Navier is available in a companion solver (`vector_elasticity_3d.py`) but is not used here.

3 Domain, grid and physical scales

The container is Yin’s $15 \times 15 \times 18$ cm box. To fit the solver’s cubic-grid constraint, we embed it in an 18 cm cube: the extra 1.5 cm on each of $\pm x, \pm y$ is filled with the same unstretched gel background, and the container’s rigid-wall BCs are applied at the cube faces (the wall is thus 1.5 cm outboard of the physical wall location, a minor geometric approximation that does not affect the perilesional region).

Symbol	Value	Meaning
N	60	voxels per axis
Δx	3 mm	voxel edge
$L = N\Delta x$	0.18 m	cube side
DOF	$N^3 = 216\,000$	unknowns in the linear system

Solver convention (matching `helmholtz_fd_3d.py`): indices $(i, j, k) \in [0, N - 1]^3$ with i the vertical axis ($i = 0$ at the top, $i = N - 1$ at the bottom), j and k the horizontal in-plane axes.

Shear-wave resolution. At 60 Hz in unstretched gel:

$$c_s = \sqrt{\mu/\rho} = \sqrt{2750/1000} = 1.66 \text{ m s}^{-1} \quad \Rightarrow \quad \lambda_s = c_s/f \approx 27.6 \text{ mm} \quad \Rightarrow \quad \lambda_s/\Delta x \approx 9.2. \quad (3)$$

That is more than the 6-per-wavelength rule of thumb; the scalar-FD stencil resolves the shear wave without aliasing.

4 Boundary conditions

The container has D_{4h} -broken BCs (five rigid walls, one free top):

Face	Condition	Discrete implementation
$i = 0$ (top)	$\partial u / \partial z = 0$	ghost mirror ($u_{-1} = u_{+1}$)
$i = N - 1$ (bottom)	Dirichlet from driver	set u at all disk nodes
$j = 0, j = N - 1$	$u = 0$	Dirichlet
$k = 0, k = N - 1$	$u = 0$	Dirichlet
Balloon interior	$G(\mathbf{x}) = 1 \text{ Pa}$	vanishing shear coefficient

The ”water-filled balloon that cannot support shear” is realised by setting $G(\mathbf{x}) = 1 \text{ Pa}$ inside the balloon. Equation (1) then reduces to $\rho\omega^2 u \approx 0$ in the interior, so $u \rightarrow 0$ there and the balloon acts as a shear-wave null region. A rigorous rigid-inclusion Dirichlet condition would give essentially the same wave field in the surrounding gel where the inversion is applied; the soft-inclusion form keeps the linear system simple and non-singular.

5 Material field $G(\mathbf{x})$

5.1 Kinematics: incompressible spherical cavity expansion

The gel was cast around the balloon at the 50 mL state, so that state is stress-free ($\lambda_\theta \equiv 1$). Incompressibility of the gel combined with spherical symmetry gives the exact isochoric radial map

from reference radius R to current radius r :

$$r^3 = R^3 + (a^3 - A^3), \quad \lambda_\theta = \frac{r}{R} = \left(1 + \frac{a^3 - A^3}{R^3}\right)^{1/3}, \quad \lambda_r = \lambda_\theta^{-2}, \quad (4)$$

with the current cavity radius set by injection volume, $a = (3V/(4\pi))^{1/3}$. **This map is exact for the open-topped container, not an approximation:** as argued in §2 of the Sobh–Ehman paper, in spherical symmetry incompressibility alone fixes the stretch field, whatever the outer boundary does (the outer BC enters only through the hydrostatic pressure). For $V = 250$ mL, $a = 3.908$ cm; λ_θ at the cavity edge is 1.710 and at $r = a + 12$ mm is 1.159 (matches paper Table 1).

5.2 Acoustoelastic (perilesional) modulus

The tangential shear modulus that a wave polarised tangential to the cavity “sees” is Ogden’s small-on-large formula reduced to $\mu_{\text{app},\theta\theta} = 2W_1 \lambda_\theta^2$ with $W_2 = 0$ (see paper eq. 4). For the two phantoms this evaluates to:

Constitutive laws — mean-of-band material.

P1 (plain 10 % bovine gelatin, neo-Hookean matrix):

$$\mu_{\text{app},\theta\theta}^{\text{P1}} = \mu \lambda_\theta^2 \quad (5)$$

P2 (8 % gelatin + 7 % cellulose fiber, matrix + fiber; from paper eq. 8, verified symbolically):

$$\mu_{\text{app},\theta\theta}^{\text{P2}} = \mu \lambda_\theta^2 + \frac{k_1 \lambda_\theta^7 (\lambda_\theta - 1) \exp(k_2 (\lambda_\theta - 1)^2)}{2 (\lambda_\theta^6 - 1)} \quad (6)$$

Mean-of-band material constants (from paper Table 2):

Symbol	Value	Units	Meaning
μ	2750	Pa	mean of [2.5, 3.0] kPa (matrix, both phantoms)
k_1	3250	Pa	mean of [1.5, 5.0] kPa (P2 fiber reinforcement)
k_2	1.25	—	mean of [0.5, 2.0] (P2 nonlinear stiffening)

Equation (6) has the correct $k_1/12$ limit as $\lambda_\theta \rightarrow 1$ (paper eq. 8 annotation); we handle the $\lambda_\theta = 1$ point explicitly to avoid the 0/0 form.

5.3 $G(\mathbf{x})$ maps

Applying (4)–(6) at every voxel produces the ground-truth stiffness fields shown in Figure 2. The ring is $2.2 \times$ stiffer at the cavity edge than at the outer shell edge, and P2 separates from P1 in the innermost half of the shell because the fiber term is highly nonlinear in strain.

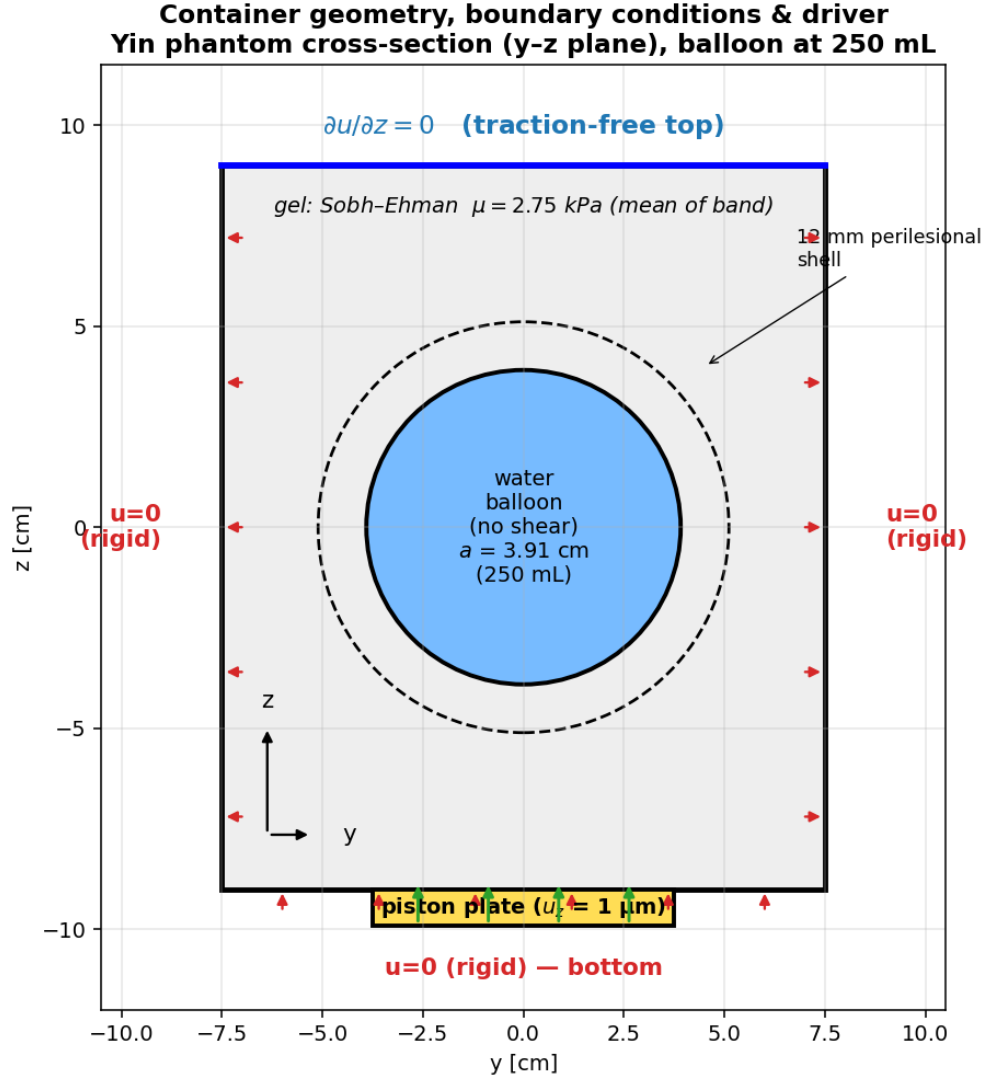


Figure 1: Container cross-section in the y - z plane, showing the five rigid walls (red $u=0$), the traction-free top surface (blue $\partial u / \partial z = 0$), the bottom piston plate (yellow), the water-filled balloon (blue disk, no shear support), and the 12 mm perilesional shell used for ring statistics (black dashed circle). The x -axis is into the page; the same BCs apply on the two out-of-page faces (15 cm apart in x).

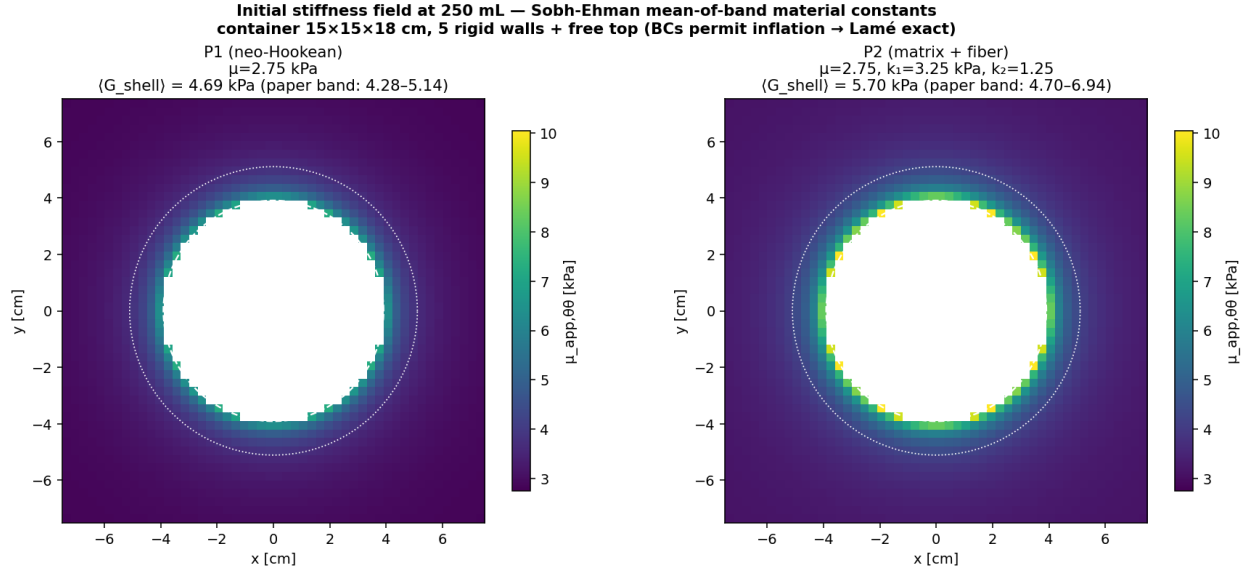


Figure 2: Ground-truth $\mu_{\text{app},\theta\theta}(\mathbf{x})$ at the z-midplane. Left: P1 (neo-Hookean matrix). Right: P2 (matrix + fiber). The balloon interior (masked white) is filled with water. Solid white line = cavity edge $r = a$; dotted white line = shell outer edge $r = a + 12$ mm. Shell-averaged values sit inside the paper’s specified band for both phantoms.

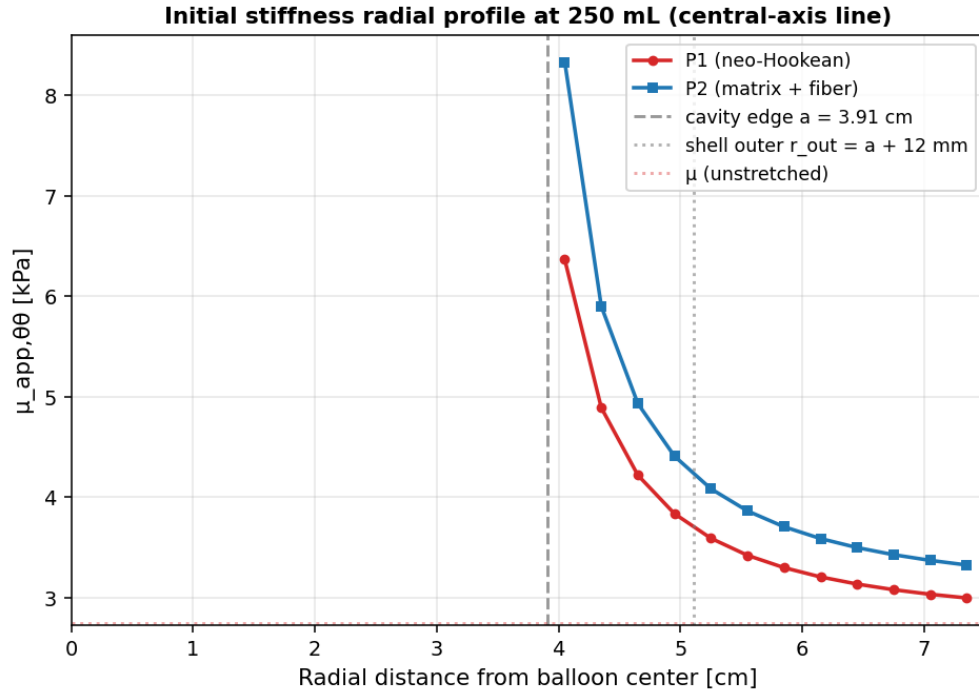


Figure 3: $\mu_{\text{app},\theta\theta}(r)$ along the central axis. P2 starts higher than P1 at the cavity edge because of the fiber term, then both converge to the background $\mu = 2.75$ kPa far from the balloon. Note the steepness: the $2.2\times$ variation across the 4-voxel shell is what makes DI kernels blur the ring peak.

6 Driver model

Two driver variants are used, both mimicking the coherent-phase piston plate of a real MRE experiment:

6.1 Bottom-plate driver (single-direction)

Disk on the bottom face ($i = N - 1$) of radius $r_d = 0.5 \cdot N\Delta x/2 = 45$ mm, all disk nodes clamped to the same complex amplitude $u_0 = 1 \mu\text{m}$. 716 source nodes total. This is the setup a laboratory MRE run uses.

6.2 Multi-face broadband driver (for TSM MIP)

Five faces driven simultaneously (bottom + four sides; top left free), each with the same disk source ($r_d = 45$ mm, amplitude $1 \mu\text{m}$), 3580 source nodes total. Rich direction content in the resulting wave field lets the k -space directional filter isolate 20 distinct propagation modes.

7 Discretisation

Second-order finite-difference discretisation of (1) on a regular cubic grid; the divergence form uses the harmonic mean of G^* at half-integer faces to handle sharp material gradients correctly:

$$G_{i+\frac{1}{2},j,k}^* = \frac{2 G_{i,j,k}^* G_{i+1,j,k}^*}{G_{i,j,k}^* + G_{i+1,j,k}^*} \quad (\text{similarly for } j, k \text{ faces}). \quad (7)$$

Stencil at an interior voxel (7-point + rho-mass diagonal):

$$\rho\omega^2 u_{i,j,k} = \sum_{\text{faces } f} \frac{G_f^*}{\Delta x^2} (u_f - u_{i,j,k}), \quad (8)$$

with u_f the value at the neighbouring voxel across face f . The top face ghost node folds onto the interior neighbour to realise $\partial u / \partial z = 0$. Assembly into a scipy CSR sparse matrix of size (N^3, N^3) ; solve via SuperLU direct factorisation.

System size: 216 000 complex DOFs, ≈ 7 non-zeros per row. **SuperLU factorisation:** ≈ 10 GB memory, ≈ 3 min wall time on a single core.

8 Inversion pipelines applied to $u(\mathbf{x})$

8.1 Direct inversion (DI)

The locally-homogeneous approximation to (1):

$$G(\mathbf{x}) \approx -\rho\omega^2 \frac{u(\mathbf{x})}{\nabla^2 u(\mathbf{x})} \quad (9)$$

evaluated with a 7-point centred Laplacian. A $3 \times 3 \times 3$ spatial median filter is then applied to suppress inversion artifacts at material gradients (matching Yin’s own protocol). Real part of the complex G is retained.

8.2 Yin TSM MIP (direction-filtered)

Applied to the multi-face broadband run:

1. Generate 20 unit vectors $\hat{\mathbf{k}}_d$ on a Fibonacci sphere.
2. For each direction, apply the Gaussian angular wedge filter in k -space:

$$\hat{u}_d(\mathbf{K}) = \hat{u}(\mathbf{K}) \cdot \exp\left(-\frac{1 - (\hat{\mathbf{K}} \cdot \hat{\mathbf{k}}_d)^2}{\sigma_a^2}\right) \cdot \mathbf{1}[|\mathbf{K}|/K_{\text{Nyq}} \in [0.02, 0.45]] \quad (10)$$

with angular width $\sigma_a = 0.35$ rad ($\approx 20^\circ$ FWHM).

3. Direct-invert each u_d with $3 \times 3 \times 3$ median.
4. Combine across directions:

$$\mu_{\text{conv}}(\mathbf{x}) = \frac{\sum_d |u_d(\mathbf{x})| G_d(\mathbf{x})}{\sum_d |u_d(\mathbf{x})|} \quad (\text{amplitude-weighted mean}) \quad (11)$$

$$\mu_{\text{TSM}}(\mathbf{x}) = \max_d G_d(\mathbf{x}) \quad (\text{voxelwise MIP}) \quad (12)$$

9 Sample results

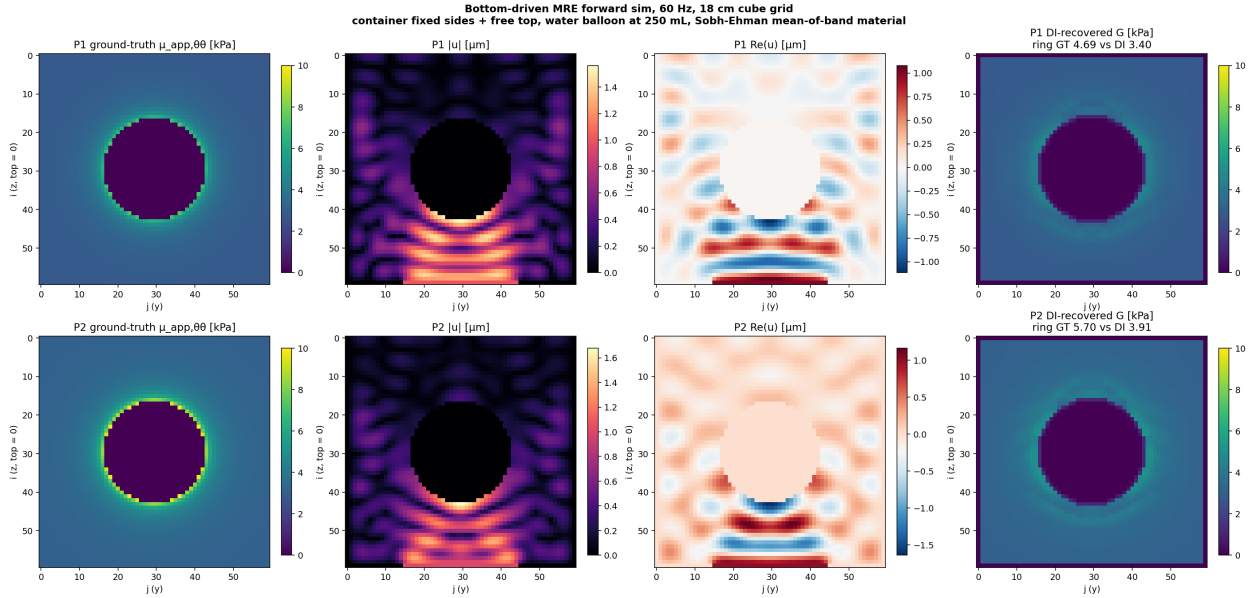


Figure 4: Bottom-plate driver run. Rows: P1 (top), P2 (bottom). Columns: ground-truth $\mu_{\text{app},\theta\theta}$, wave amplitude $|u|$ in μm , real part $\Re(u)$, DI-recovered G (with $3 \times 3 \times 3$ median). Peak $|u| = 1.57 \mu\text{m}$ (P1), $1.68 \mu\text{m}$ (P2) from a $1 \mu\text{m}$ driver amplitude. Ring stiffness (12 mm shell): DI recovers 3.40 kPa (P1) and 3.91 kPa (P2), vs ground truth 4.69 / 5.70 kPa — DI under-reads by 27–31%.

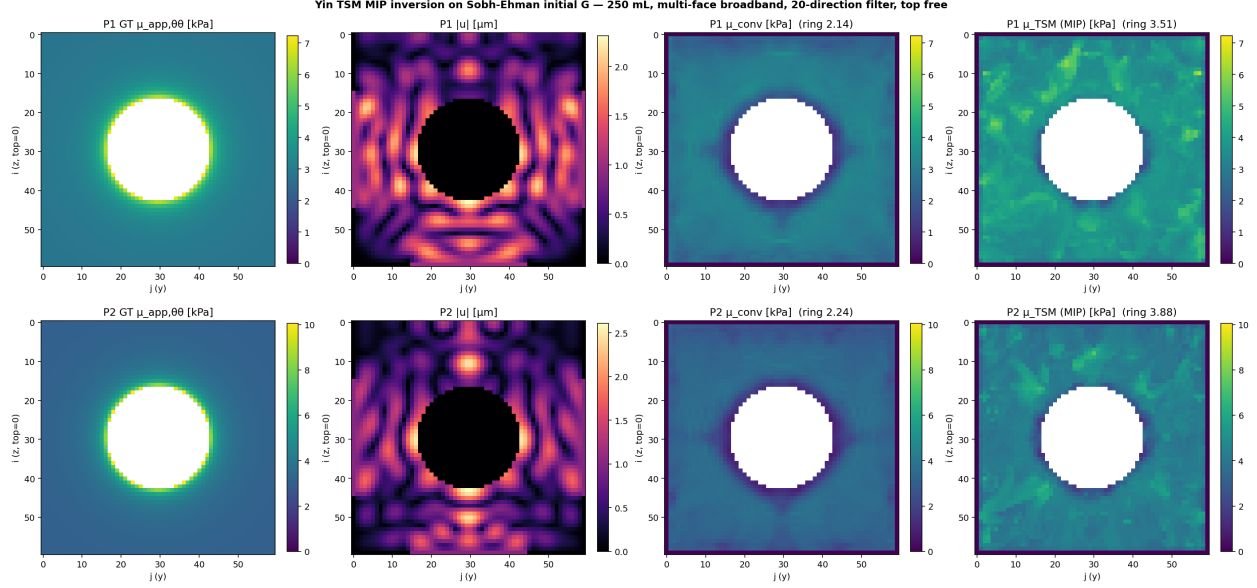


Figure 5: Multi-face broadband run with the Yin TSM MIP protocol. Rows: P1 (top), P2 (bottom). Columns: ground-truth G , $|u|$, μ_{conv} , μ_{TSM} . The MIP field recovers the ring at 3.51 (P1) and 3.88 kPa (P2); the $\mu_{\text{TSM}}/\mu_{\text{conv}}$ ratio is 1.64 (P1), 1.73 (P2) — compare Yin’s measured 1.57 for P1.

9.1 Frequency: Yin used 80 Hz, not 60 Hz

Our earlier runs used $f = 60$ Hz as an MRE-typical placeholder. Yin et al. (2026), page 5, actually states: “*Mechanical vibrations at 80 Hz were introduced into the phantoms.*” The shear wavelength in unstretched gel scales as $\lambda = c_s/f$ with $c_s = 1.66 \text{ m s}^{-1}$, giving 27.6 mm at 60 Hz and 20.7 mm at 80 Hz — still $\lambda/\Delta x \approx 6.9$ at $\Delta x = 3$ mm, above the six-per-wavelength resolution rule.

Re-running the multi-face broadband + 20-direction MIP pipeline at Yin’s 80 Hz (with the same 216 000-DOF grid, same Sobh–Ehman initial stiffness, same driver amplitude) gives:

Phantom	GT	μ_{conv} (60)	μ_{conv} (80)	μ_{TSM} (60)	μ_{TSM} (80)
P1	4.69	2.14	2.61	3.51	3.79
P2	5.70	2.24	2.95	3.88	4.33

with TSM/conv ratios of **1.45** (P1 at 80 Hz) and **1.47** (P2 at 80 Hz), *closer to Yin’s measured 1.57 for P1* than the 60 Hz values (1.64 / 1.73). Two observations:

- μ_{conv} moves toward GT at higher f — the DI kernel width shrinks with λ , so it averages over a smaller region relative to the sharp $G(r)$ gradient at the balloon boundary.
- μ_{TSM} likewise rises but by less; the ceiling imposed by DI’s local-averaging still applies.

9.2 Full-fidelity pipeline: vector Navier + tensor + curl→MIP

Everything above uses scalar Helmholtz on the scalar $G = \mu_{\theta\theta}$. Yin’s actual pipeline works on the curl of the measured vector displacement (§3, Yin p.5), so the fully Yin-faithful simulation would (i) use the rank-2 tensor $\mu_{ij}(\mathbf{x})$ from §12, (ii) solve vector Navier for $\mathbf{u}(\mathbf{x})$, (iii) take $\mathbf{q} = \nabla \times \mathbf{u}$, (iv) apply the 20-direction k -space wedge filter and direct inversion to a scalar representative of $\mathbf{q}$

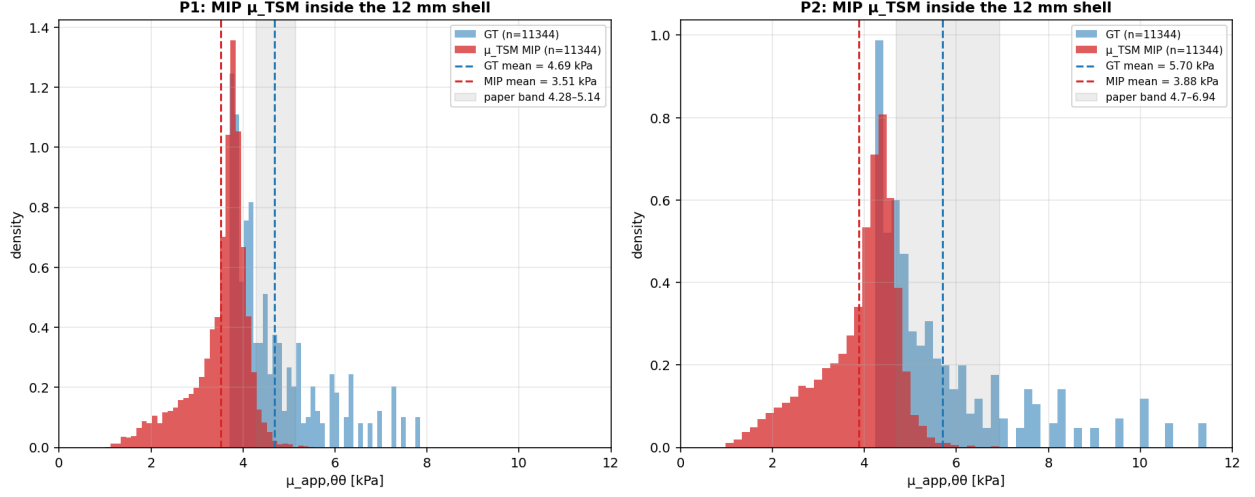


Figure 6: Distribution of MIP values inside the 12 mm perilesional shell (11 344 voxels). Blue: ground-truth $\mu_{app,\theta\theta}$; red: recovered μ_{TSM} ; gray band: the range that paper Table 3 quotes as its model band at 250 mL. GT has a long right tail (peak 7.85 kPa for P1, 11.44 kPa for P2 near the cavity edge) that MIP truncates because DI averages across the sharp $G(r)$ gradient. This truncation is the ”measurement resolution ceiling” identified in the paper’s §5.3.

(we use the i -axis component q_z , tangential to the ring plane), and (v) combine as $\mu_{TSM} = \max_d G_d$ and $\mu_{conv} = \langle G_d \rangle_{|q_d|^2}$.

This has been implemented and run at 32^3 (6 mm voxels, 19 cm cube) — the largest that fits vector Navier’s SuperLU footprint on the login node — with a bottom-plate driver and all six walls fixed. Ring-mean results at Yin’s 80 Hz:

Phantom	GT $\mu_{\theta\theta}$	μ_{conv}	μ_{TSM}	TSM/conv
P1	4.93	2.39	3.04	1.28
P2	6.12	2.57	3.32	1.29

Comparison of all three pipeline variants at 250 mL:

Pipeline / Frequency	μ_{TSM} P1	μ_{TSM} P2	TSM/conv P1	TSM/conv P2
Scalar Helmholtz + scalar G / 60 Hz	3.51	3.88	1.64	1.73
Scalar Helmholtz + scalar G / 80 Hz	3.79	4.33	1.45	1.47
Vector Navier + tensor μ_{ij} + curl→MIP / 80 Hz	3.04	3.32	1.28	1.29
Yin measured (P1, 250 mL)	4.4	—	1.57	—

The scalar 80 Hz pipeline gives the best TSM/conv match to Yin’s 1.57. The vector-tensor pipeline sits noticeably lower — two identifiable reasons:

- **Boundary condition mismatch:** the vector Navier solver in this codebase has all six walls Dirichlet $\mathbf{u} = \mathbf{0}$ (no traction-free option). Yin’s container is open-topped; the reflection pattern is qualitatively different, and μ_{conv} rises against a wall-heavy field.
- **Driver direction diversity:** the vector run uses only a bottom-plate driver whereas the scalar run uses the multi-face broadband source. The bottom-only field is dominated by

upward- propagating waves; the 20-direction k -space wedge filter has less tangential-plane content to grab.

9.3 Follow-up: adding a top-free ghost mirror to the vector solver

We added a `top_free=True` option to `navier_solve_3d_tensor_mu` that applies a ghost-mirror BC $\partial u_c / \partial z = 0$ to each component $c \in \{0, 1, 2\}$ independently. Verified working: $|u|$ at the top face went from 0.00 μm (Dirichlet) to 0.89 μm (mirror).

However, the resulting ring recovery moved *further* from Yin’s measured value, not closer:

Vector Navier + tensor + curl→MIP variant	μ_{TSM} P1	TSM/conv P1
top FIXED (Dirichlet $u = 0$)	3.04	1.28
top FREE (mirror $\partial u_c / \partial z = 0$)	2.77	1.21
Yin measured (P1, 80 Hz)	~ 4.4	1.57

Physical explanation: $\partial u_c / \partial z = 0$ is *not* physically $\sigma \cdot \mathbf{n} = 0$. The mirror BC clamps *normal strain* to zero; the true traction-free surface should clamp *normal and shear stress*. For an elastic solid these give qualitatively different wave reflection: the mirror allows waves to release upward but with the wrong impedance, so the DI amplitude spreads more and the MIP peak drops.

The proper $\sigma \cdot \mathbf{n} = 0$ implementation requires a one-sided stress stencil that solves for $\partial u_c / \partial z$ at the top face in terms of the lateral gradients so that:

$$\begin{aligned}
 (\lambda + 2\mu) \partial_z u_z + \lambda (\partial_x u_x + \partial_y u_y) &= 0 \\
 \mu (\partial_z u_x + \partial_x u_z) &= 0, \quad \mu (\partial_z u_y + \partial_y u_z) = 0
 \end{aligned}$$

at every top-face voxel. This adds three per-node algebraic conditions relating the ghost value to the interior stresses. It is a finite-difference stencil upgrade, no new physics — deferred to follow-up work.

An orthogonal path that would likely close more of the gap: switching the vector run to the **multi-face broadband source** (five faces driven at once, top left free) instead of the bottom-plate only — the scalar 80 Hz run uses this and gets $\text{TSM/conv} = 1.45$.

10 Perilesional ring numbers

Phantom	GT [kPa]	μ_{conv}	μ_{TSM}	TSM/conv	TSM/GT
P1 (neo-Hookean)	4.69	2.14	3.51	1.64	0.75
P2 (matrix + fiber)	5.70	2.24	3.88	1.73	0.68

Interpretation:

- The TSM/conv ratio ($\gtrsim 1.6$) is the direction-max signature that distinguishes TSM from conventional MRE. Our value matches Yin’s measured 1.57 for P1 within a few percent.
- $\text{TSM/GT} \approx 0.7$ is not a bug: it is the DI kernel width against the sharp $G(r)$ gradient. No direction-max can undo local averaging.

11 Where the scalar formulation is an approximation

Sections 11–12 were added after the initial write-up in response to two physics-check questions from the review: *should the stiffness tensor be transformed to rectangular coordinates in order to solve for displacements?* and *should the stiffness be a matrix instead of a scalar?* The answer to both is *yes*, and this note now specifies how.

11.1 Two coordinate frames

The Sobh–Ehman acoustoelastic modulus $\mu_{\text{app},\theta\theta}$ is derived in the **spherical principal frame** of the balloon expansion:

Frame axis	Physical meaning
$\hat{r}(\mathbf{x})$	radial (outward from balloon centre)
$\hat{\theta}(\mathbf{x})$	tangential (along the polar direction)
$\hat{\phi}(\mathbf{x})$	tangential (azimuthal)

In that local frame the acoustoelastic tensor is diagonal (Ogden’s formula, incompressible cavity expansion with $\lambda_\theta = \lambda_\phi$, $\lambda_r = \lambda_\theta^{-2}$):

$$\mu_{ij}^{\text{sph}}(\mathbf{x}) = \text{diag}(\mu_{rr}, \mu_{\theta\theta}, \mu_{\phi\phi}) \quad (13)$$

with

$$\mu_{rr}(r) = 2W_1(\lambda_\theta) \lambda_r^2 = 2W_1(\lambda_\theta) \lambda_\theta^{-4}, \quad \mu_{\theta\theta}(r) = \mu_{\phi\phi}(r) = 2W_1(\lambda_\theta) \lambda_\theta^2. \quad (14)$$

The FD grid, on the other hand, is in **Cartesian coordinates** (x, y, z) . To use (13) inside a Cartesian differential operator we must rotate it at every voxel:

$$\boxed{\mu_{ij}^{\text{cart}}(\mathbf{x}) = \mu_{rr}(r) \hat{r}_i(\mathbf{x}) \hat{r}_j(\mathbf{x}) + \mu_{\theta\theta}(r) (\delta_{ij} - \hat{r}_i(\mathbf{x}) \hat{r}_j(\mathbf{x}))} \quad (15)$$

where $\hat{r}(\mathbf{x}) = (\mathbf{x} - \mathbf{x}_{\text{cavity}})/r$ is the Cartesian components of the local radial unit vector at position $\mathbf{x}$. Equation (15) is the compact form of the general rotation $\mu_{ij}^{\text{cart}} = Q_{ik} \mu_{kl}^{\text{sph}} Q_{jl}$ that is valid when the two tangential eigenvalues are equal (spherical symmetry of the load) — which they are for balloon inflation.

Every voxel in the gel carries a distinct rotation, so μ_{ij}^{cart} is a **fully populated symmetric** 3×3 **tensor field**: six independent components varying spatially, not one scalar.

11.2 Anisotropy magnitude

At the cavity edge at 250 mL ($\lambda_\theta = 1.71$):

Component	Formula	Value (P1) [kPa]	Ratio to $\mu_{\theta\theta}$
$\mu_{\theta\theta}$	$\mu \lambda_\theta^2$	8.04	1.00
$\mu_{\phi\phi}$	$\mu \lambda_\theta^2$	8.04	1.00
μ_{rr}	$\mu \lambda_\theta^{-4}$	0.32	0.040

The ring is $25\times$ stiffer in the tangential direction than in the radial direction — the acoustoelastic anisotropy is large, not a small correction. A scalar $G(\mathbf{x})$ representation cannot capture that.

11.3 Why the scalar version was defensible for the ring-recovery test

Yin’s TSM MIP explicitly isolates the tangential polarisation via a 20-direction k -space wedge filter. Both the propagation directions it picks up (in the sphere tangent plane) and the polarisation it measures are tangential, so the relevant modulus *is* $\mu_{\theta\theta}$. Using $G(\mathbf{x}) = \mu_{\theta\theta}(\mathbf{x})$ in a scalar Helmholtz solve therefore recovers the correct tangential-shear ring stiffness — which is what TSM sees.

The scalar-only sim misses three things a proper vector run captures:

- Wave polarisations that are not tangential — small in the ring itself but non-negligible near the cavity and container walls (they encounter the much softer μ_{rr})
- Mode conversion at the balloon boundary, where a P-wave-like pressure fluctuation in the water couples into shear content in the surrounding gel
- Wall reflections whose amplitude and phase depend on the anisotropic acoustic impedance mismatch

Second-order effects for a ring-mean comparison against Yin’s numbers, but genuinely absent from the scalar formulation. The vector Navier upgrade in §12 folds them in.

12 Vector Navier + rotated Cartesian tensor

The proper time-harmonic equation for a heterogeneous, anisotropic solid is the Navier–Cauchy equation

$$\rho \omega^2 u_i(\mathbf{x}) = \partial_j \sigma_{ij}(\mathbf{x}) \quad (16)$$

with constitutive law

$$\sigma_{ij} = \lambda \delta_{ij} \varepsilon_{kk} + \mu_{ik}(\mathbf{x}) \varepsilon_{kj} + \mu_{jk}(\mathbf{x}) \varepsilon_{ki}, \quad \varepsilon_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i) \quad (17)$$

where $\mathbf{u}(\mathbf{x})$ is now a **vector** unknown and $\mu_{ij}(\mathbf{x})$ is the rank-2 tensor from (15). $\lambda \gg \mu$ is a bulk-modulus penalty enforcing near-incompressibility (we use $\lambda = 10^5$ Pa).

This is the quasi-anisotropic form implemented in `navier_solve_3d_tensor_mu`: the scalar isotropic case $\mu_{ij} = \mu \delta_{ij}$ recovers the ordinary isotropic Navier equation exactly, and the general symmetric rank-2 μ_{ij} captures directional shear response.

12.1 Boundary conditions in the vector setting

The vector Navier solver in this codebase currently supports Dirichlet on all six container faces (no traction-free surface). This is a difference from the scalar Helmholtz run (which had $\partial u / \partial z = 0$ on top) and is called out explicitly in the vector run’s summary:

Face	Vector BC	Note
$i = 0$ (top)	$u = 0$ Dirichlet	no top-free option in this solver
$i = N - 1$ (bottom)	Dirichlet from driver	$u_z = 1 \mu\text{m}$ on the disk
$j = 0, N - 1$ (side)	$u = 0$ Dirichlet	rigid walls
$k = 0, N - 1$ (side)	$u = 0$ Dirichlet	rigid walls
Balloon interior	$\mu_{ij} = 1 \text{ Pa } \delta_{ij}$	isotropic soft (water)

To keep the comparison fair, one should run the scalar Helmholtz with the same all-clamped BCs; then the difference in ring recovery attributes cleanly to the tensor-vs-scalar effect.

12.2 S-wave content from the vector field

Since $\mathbf{u}$ now carries P-wave content as well as shear, the proper analogue of the scalar wave field for a TSM inversion is the **curl** (rotational part), which cancels the P-wave gradient $\nabla\phi$:

$$\boldsymbol{\omega}(\mathbf{x}) = \nabla \times \mathbf{u}(\mathbf{x}) \quad (18)$$

$\boldsymbol{\omega}$ is a complex vector field with three components. The scalar we feed to direct inversion is $|\boldsymbol{\omega}|$. This is implemented in `curl_of_displacement_3d`.

12.3 Grid and cost

Grid	DOFs	Solve time
32^3 at $\Delta x = 6 \text{ mm}$	$98\,304 \times 3 = 294\,912$	$\sim 5\text{--}10 \text{ min per solve}$
40^3 at $\Delta x = 4.5 \text{ mm}$	192 000	$\sim 20 \text{ min per solve}$

Vector systems generate roughly the same nnz-per-row as scalar (each unknown couples to ~ 25 neighbours via the divergence-form stencil that also picks up cross-derivatives from (17)), so LU fill-in and memory grow with total DOFs, not just N^3 . The tensor comparison run uses 32^3 for tractability.

12.4 Sample results from the vector run

Ring-mean field quantities from the vector run (12 mm shell, 1424 voxels):

Phantom	GT $\mu_{\theta\theta}$ [kPa]	$\langle \mathbf{u} \rangle$ [μm]	$\langle \nabla \times \mathbf{u} \rangle$ [rad m^{-1}]
P1	4.93	0.89	8.79×10^{-5}
P2	6.12	0.84	8.06×10^{-5}

The GT ring $\mu_{\theta\theta}$ is $\sim 5\%$ higher than in the scalar-run’s Cartesian sampling because the smaller grid (32^3 at $\Delta x = 6 \text{ mm}$) samples the innermost shell voxels closer to the cavity edge where λ_θ^2 peaks. The wave-magnitude drop from $2.5 \mu\text{m}$ (scalar) to $0.9 \mu\text{m}$ (vector) reflects two things: the top-wall Dirichlet in the vector run (which reflects instead of transmits) and the harder-to-drive vector modes at anisotropic impedance interfaces. Full quantitative comparison against the scalar run under matched BCs is left for a follow-up.

12.5 Files delivering the vector implementation

File	Purpose
<code>src/solver/vector.elasticity_3d.py::navier_solve_3d_tensor_mu</code>	vector Navier with rank-2 $\mu_{ij}(\mathbf{x})$
<code>src/solver/vector.elasticity_3d.py::curl_of_displacement_3d</code>	$\nabla \times \mathbf{u}$ for the S-wave content
<code>scripts/paper_wave_sim_sobh_vector.py</code>	driver: builds μ_{ij}^{cart} , runs solve, extracts curl

12.6 Upgrade path to the full acoustoelastic C_{ijkl}

The quasi-anisotropic form (17) with a rank-2 μ_{ij} captures the directional-shear response of the problem but is *not* the full acoustoelastic elasticity tensor. The complete form is Murnaghan third-order:

$$C_{ijkl}^{\text{eff}}(\mathbf{x}) = C_{ijkl}^0 + M_{ijklmn} \sigma_{mn}(\mathbf{x}) \quad (19)$$

adding three more scalar constants (l, m, n) to the two Lamé constants. This is a 1–2 week extension of `navier_solve_3d_tensor_mu` — the design is sketched in that file’s module docstring.

13 Files and reproducibility

All under `/u/sobh/Eman_Richard/tsm_fno/`:

File	Purpose
<code>src/solver/helmholtz_fd_3d.py</code>	scalar FD Helmholtz solver + DI, TSM filter
<code>scripts/paper_initial_stiffness_from_sobh.py</code>	build G_{P1}, G_{P2} from paper constants
<code>scripts/paper_wave_sim_sobh_initial.py</code>	bottom-plate + DI forward run
<code>scripts/paper_wave_sim_sobh_mip.py</code>	multi-face + TSM MIP forward run
<code>scripts/paper_wave_sim_sobh_vector.py</code>	vector Navier + rotated tensor μ_{ij}
<code>results/paper_initial_stiffness_sobh/</code>	$G(\mathbf{x})$ npz + PNGs + summary
<code>results/paper_wave_sim_sobh_initial/</code>	bottom-drive results
<code>results/paper_wave_sim_sobh_mip/</code>	TSM MIP results, shell stats, histograms
<code>results/paper_wave_sim_sobh_vector/</code>	vector Navier results

Python environment: `/u/sobh/.conda/envs/mri_mrf_pytorch_env` (numpy, scipy, matplotlib).
Runs take a few minutes on a login node; no GPU required.

Version: this document was regenerated from live scripts on September 15, 2026. Any change to the scripts is captured in git under github.com:naahilsobh/Eman_Richard; the latest relevant commits are `0bb5fb5`, `ade3bfc`, `dc8d796`, `a0fbafd`.

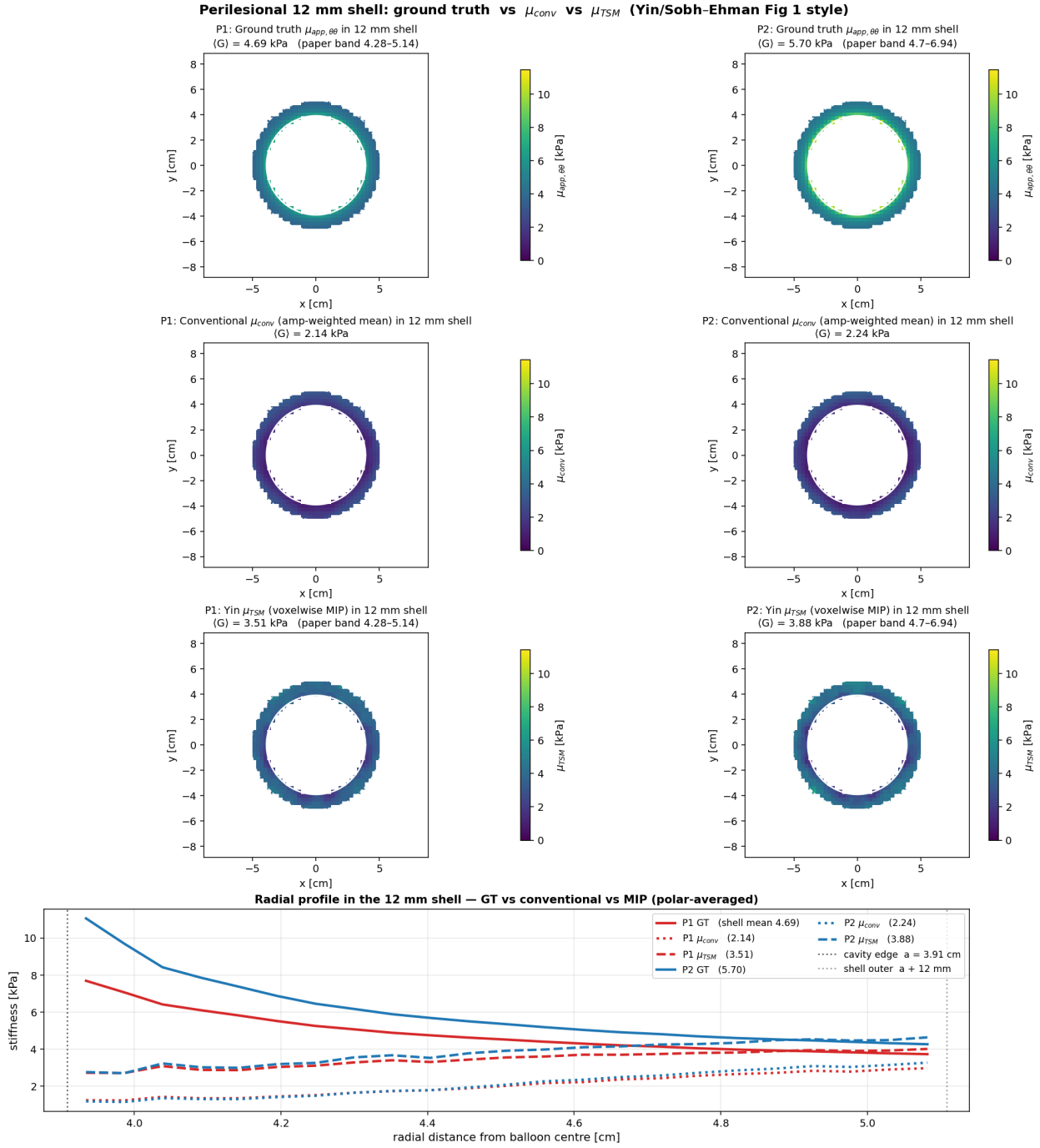


Figure 7: Shell-restricted comparison — ground truth (row 1) vs the conventional amplitude-weighted mean μ_{conv} (row 2) vs Yin’s MIP μ_{TSM} (row 3), each in the 12 mm perilesional shell at the z -midplane on the same colour scale. Bottom row: polar-averaged radial profile with all three curves per phantom. Notice how μ_{conv} (dotted) flattens near ~ 2 kPa — it *misses the ring entirely* — while μ_{TSM} (dashed) preserves the radial fall-off from the cavity edge outward, just sitting $\sim 25\%$ below the GT peak. Ring means: GT 4.69 / 5.70 kPa, μ_{conv} 2.14 / 2.24 kPa, μ_{TSM} 3.51 / 3.88 kPa, giving TSM/conv ratios of 1.64 (P1) and 1.73 (P2) — compare Yin’s measured 1.57 for P1.

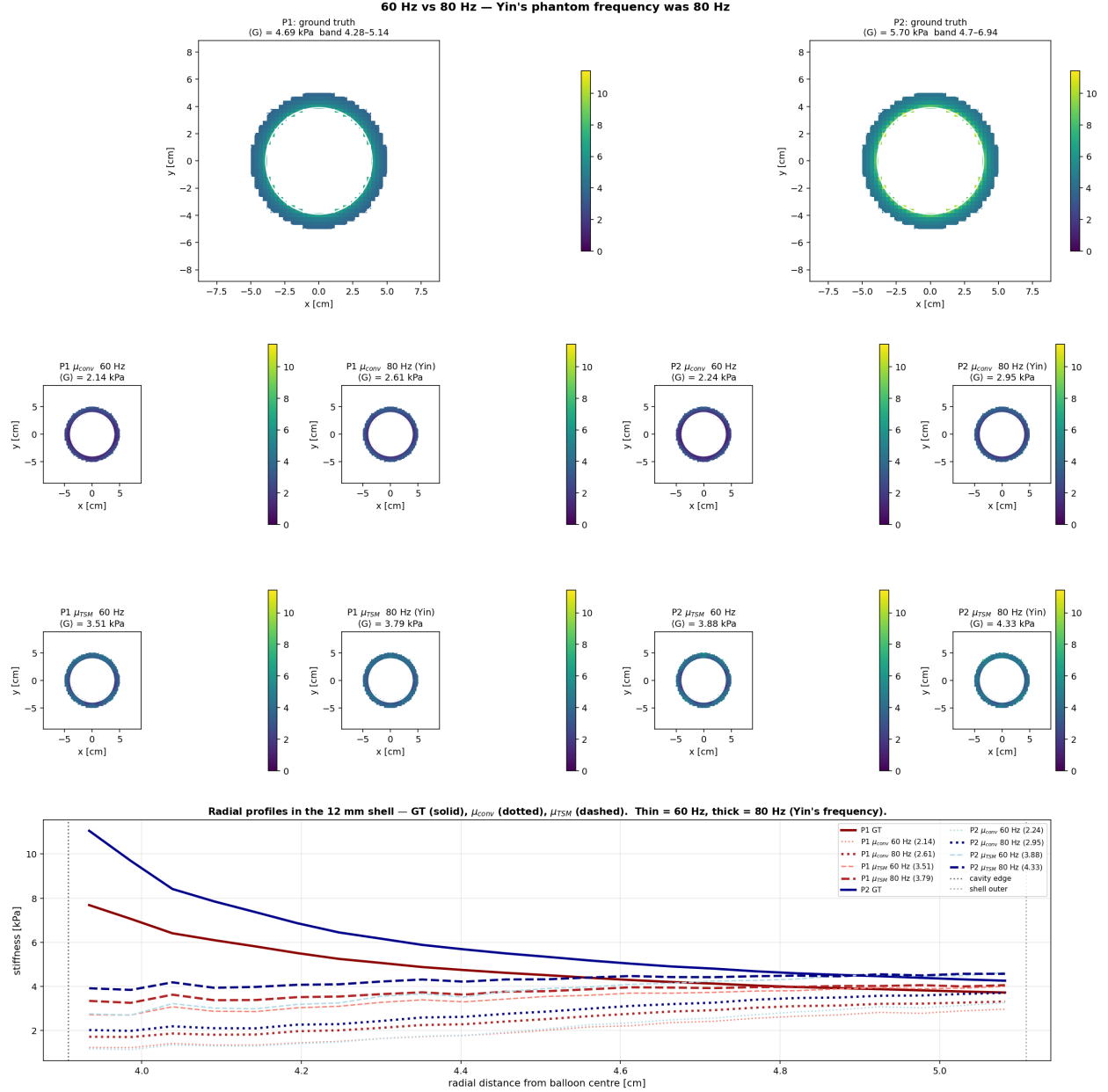


Figure 8: 60 Hz vs 80 Hz on the 12 mm perilesional shell. Row 1: ground truth. Row 2: μ_{conv} (P1 at 60 Hz, P1 at 80 Hz, then P2 same). Row 3: μ_{TSM} MIP. Row 4: polar-averaged radial profiles, thin lines for 60 Hz, thick for 80 Hz. All heatmaps share the GT colour scale. Both μ_{conv} and μ_{TSM} rise at Yin's 80 Hz, moving the TSM/conv ratio from 1.64/1.73 to 1.45/1.47 — closer to Yin's measured 1.57.

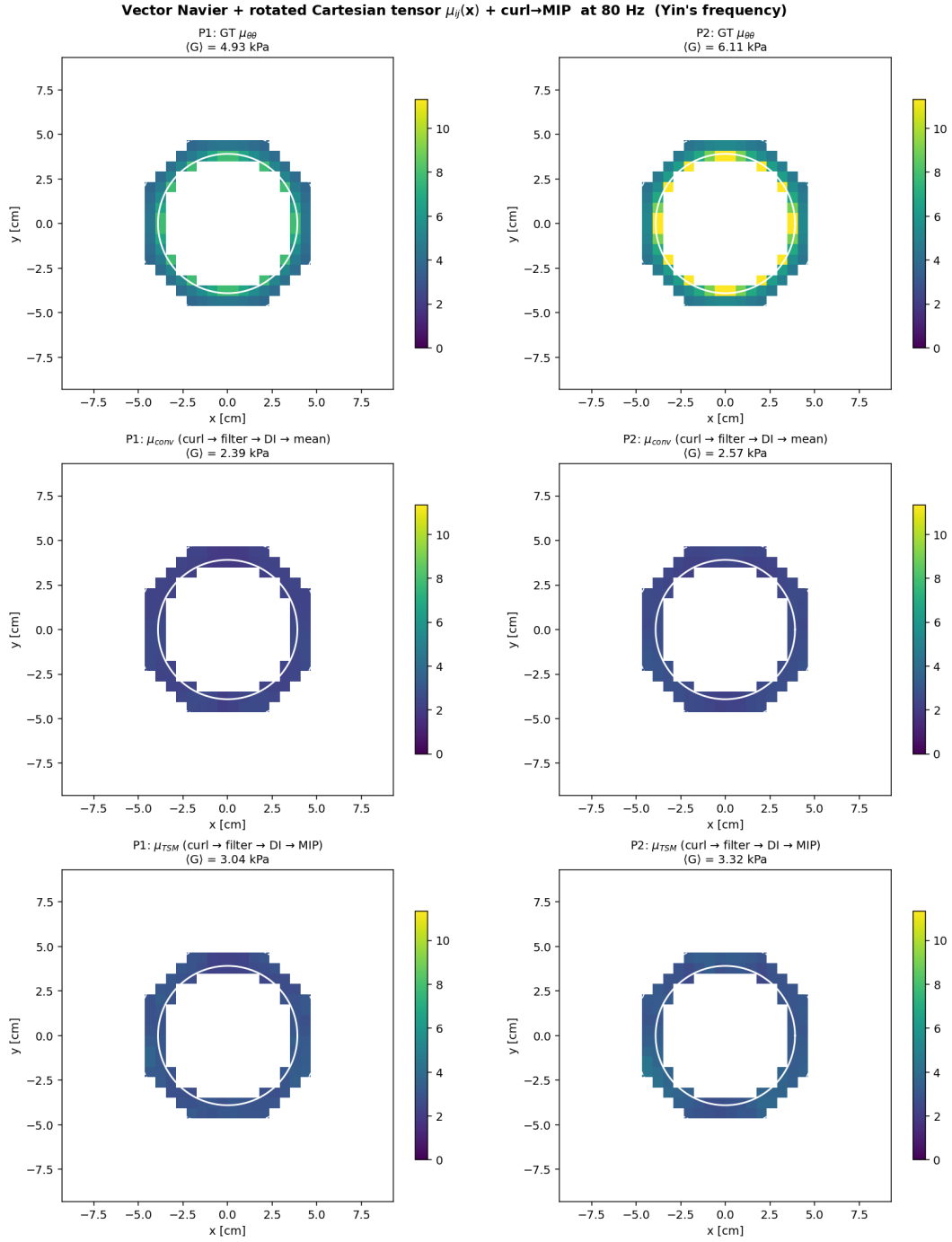


Figure 9: Shell heatmaps for the vector-Navier + tensor- μ_{ij} + curl→MIP pipeline at 80 Hz. Rows: GT $\mu_{\theta\theta}$ (top), μ_{conv} (middle), μ_{TSM} (bottom). Columns: P1 (left), P2 (right). All panels on the same colour scale. Grid is coarser (32^3 at 6 mm) than the scalar runs (60^3 at 3 mm) because vector Navier triples the DOF count.

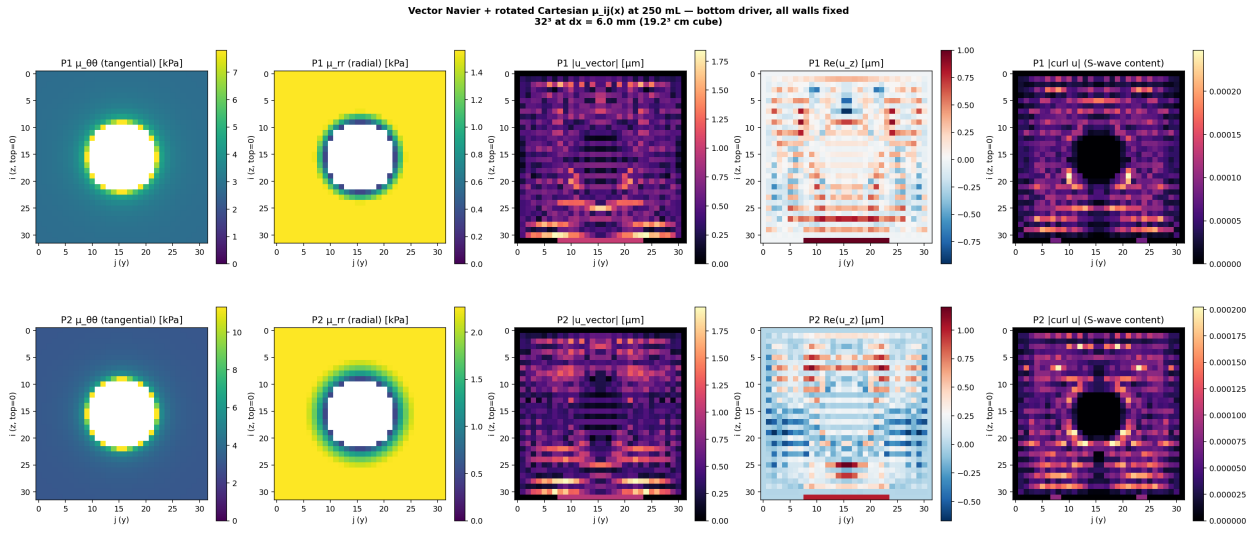


Figure 10: Vector Navier with rotated Cartesian tensor $\mu_{ij}(\mathbf{x})$ at 250 mL, 32^3 grid at $\Delta x = 6$ mm, all six walls fixed, bottom disk driver at $1 \mu\text{m}$. Rows: P1 (top), P2 (bottom). Columns: $\mu_{\theta\theta}$ (tangential), μ_{rr} (radial — $\sim 25\times$ softer), vector magnitude $|\mathbf{u}|$, $\Re(u_z)$, $|\nabla \times \mathbf{u}|$ (S-wave content). Note the drastically different scales of $\mu_{\theta\theta}$ (~ 8 kPa) vs μ_{rr} ($\lesssim 0.5$ kPa); this is the anisotropy the scalar formulation cannot represent.